SPORT CONNECT

Find your team, play your sport

Inspiration Lab

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THE PROBLEM

Move to a new city, **lose** your sports community

Meetup, Sportlerei: too broad, or stuck to one country

Finding people = WhatsApp groups, random Facebook events, asking around

International students said this was their biggest social problem

OUR SOLUTION

A focused app that does one thing well: matching people to activities nearby

Local-first, built around our city and campus

Create or join an activity in seconds, chat with the group, keep playing

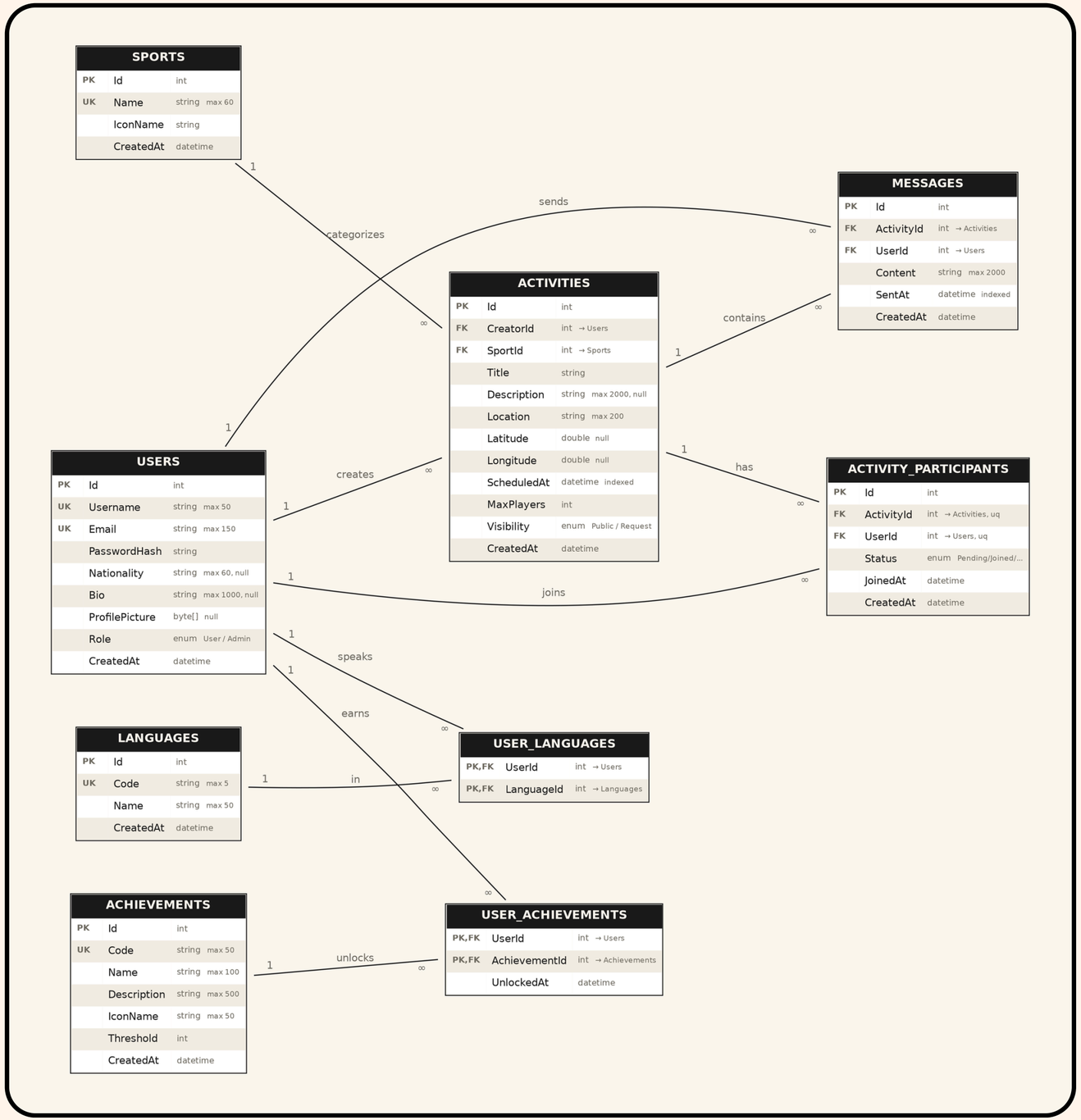
Made for newcomers and international students first

A WINDOWS DESKTOP APP WHERE YOU CAN…

- Sign up, log in, set your sports and languages
- Browse activities, filter by sport
- Create your own (place, time, max players)
- Join, leave, chat with the group
- See your stats, unlock achievements
- Admin tools for managing users

.NET 8 · WPF · EF Core 8 (SQLite) · MVVM · xUnit

ERD


SPORTS
PK Id int
UK Name string max 60
IconName string
CreatedAt datetime
MESSAGES
PK Id int
FK ActivityId int → Activities
FK UserId int → Users
Content string max 2000
SentAt datetime indexed
CreatedAt datetime
ACTIVITIES
PK Id int
FK CreatorId int → Users
FK SportId int → Sports
Title string
Description string max 2000, null
Location string max 200
Latitude double null
Longitude double null
ScheduledAt datetime indexed
MaxPlayers int
Visibility enum Public / Request
CreatedAt datetime
USERS
PK Id int
UK Username string max 50
UK Email string max 150
PasswordHash string
Nationality string max 60, null
Bio string max 1000, null
ProfilePicture byte[] null
Role enum User / Admin
CreatedAt datetime
ACTIVITY_PARTICIPANTS
PK Id int
FK ActivityId int → Activities, uq
FK UserId int → Users, uq
Status enum Pending/Joined/...
JoinedAt datetime
CreatedAt datetime
LANGUAGES
PK Id int
UK Code string max 5
Name string max 50
CreatedAt datetime
USER_LANGUAGES
PK,FK UserId int → Users
PK,FK LanguageId int → Languages
ACHIEVEMENTS
PK Id int
UK Code string max 50
Name string max 100
Description string max 500
IconName string max 50
Threshold int
CreatedAt datetime
USER_ACHIEVEMENTS
PK,FK UserId int → Users
PK,FK AchievementId int → Achievements
UnlockedAt datetime
categorizes
sends
contains
creates
has
joins
speaks
earns
in
unlocks
1
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ARCHITECTURE

1. UI → WPF + MVVM (uses Services)
2. Services → business logic (uses Data)
3. Data → EF Core (uses Domain)
4. Domain → entities + enums (no dependencies)
5. Tests → unit tests (use Services + Domain)

ARCHITECTURE & DESIGN DECISIONS

MVVM

View models know nothing about WPF windows.

Dependency Injection

Every service handed its dependencies through the constructor, easy to mock.

Repository

We use EF Core's DbSets directly instead of wrapping them.

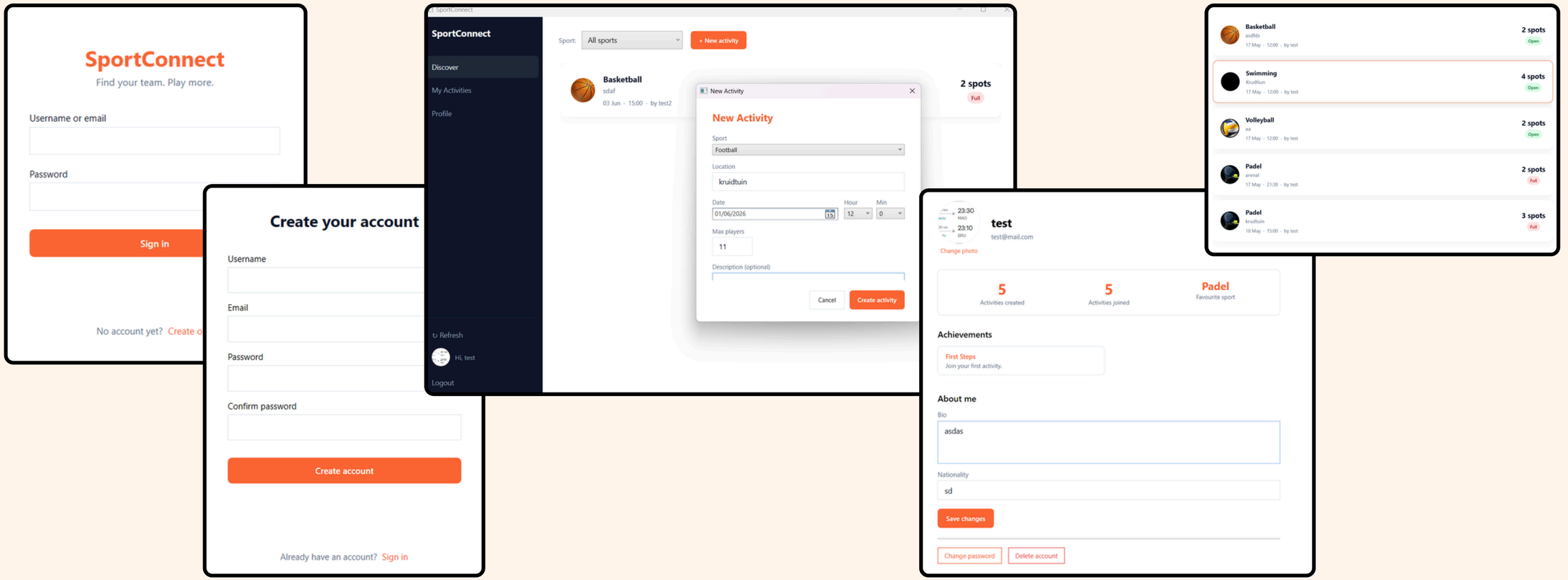
Unit of Work

Each service decides its own transaction with SaveChangesAsync.

Observer

View models raise events like LoginSucceeded and the view reacts.

OUR APP


SportConnect
Find your team. Play more.
Username or email
Password
Sign in
No account yet? Create o
Create your account
Username
Email
Password
Confirm password
Create account
Already have an account? Sign in
SportConnect
Discover
My Activities
Profile
Refresh
Hi, test
Logout
Sport: All sports
+ New activity
Basketball
sdaf
03 Jun · 15:00 · by test2
2 spots
Full
New Activity
New Activity
Sport
Football
Location
kruidtuin
Date
01/06/2026
Hour
12
Min
0
Max players
11
Description (optional)
Cancel
Create activity
Basketball
asdfds
17 May · 12:00 · by test
2 spots
Open
Swimming
Kruidtuin
17 May · 12:00 · by test
4 spots
Open
Volleyball
aa
17 May · 12:00 · by test
2 spots
Open
Padel
arenal
17 May · 21:30 · by test
2 spots
Full
Padel
kruidtuin
18 May · 15:00 · by test
3 spots
Full
23:30 MAD
23:10 BRU
test
test@mail.com
Change photo
5
Activities created
5
Activities joined
Padel
Favourite sport
Achievements
First Steps
Join your first activity.
About me
Bio
asdas
Nationality
sd
Save changes
Change password
Delete account

STRATEGY PATTERN

Every achievement is a rule behind one interface: IAchievementRule

Four rules: First Steps, Social Butterfly, Master Creator, Dedicated Player

The service doesn't know any rule by name, it gets them all injected and loops through

New achievement = one class + one line to register it. Nothing else changes.

TECH CHOICES

SQLite over SQL Server → single-user app, no server, ships with the app.

EF Core code-first → C# entities define the schema.

BCrypt → passwords never stored in plain text, slow on purpose to resist brute-force.

CommunityToolkit.Mvvm → source generators cut the boilerplate.

xUnit + EF InMemory → tests run in milliseconds.

TESTING

We tested the business logic, where bugs actually hurt:

Domain → IsFull, IsChatActive

AuthService → register, login, the bad cases

ActivityService → auto-join, capacity, leaving

AchievementService → rules fire, no duplicate unlocks

StatsService → empty user, most-played sport

Each test runs on a fresh in-memory database with a random name, so no test affects another. We tested the UI by hand, not automated.

LIVE DEMO

Register

Login, land on the feed.

Activities

Create an activity, join from another profile.

Profile

Check user’s profile and its functionalities

Admin

Enter the admin account, check everything from there

CHALLENGES

WPF + MVVM

Keeping logic out of the code-behind took discipline; events from view model to view solved it.

Chat lifecycle

A chat stays open from creation until midnight of the activity day, a rule we put on the entity itself.

Achievement timing

We stopped checking on every join and now check lazily when the profile opens. Small delay, far fewer database calls.

FUTURE WORK

Future possible updates

- Real-time chat with SignalR instead of polling
- A mobile app (the Domain layer is already reusable)
- A map view (we already store coordinates)
- Buddy-matching by shared language and sport
- Push notifications when someone joins

THANK YOU

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